‘What if?’ questions

Vary a condition to deepen understanding

Use What If questions to deepen a secure example by changing one condition. Students should adapt the reasoning they have established rather than begin an unrelated problem.

How to do it well

1 Establish a secure example. Check the class can complete and explain the basic method or idea.

2 Inspect the structure. Identify the steps or criteria that matter, using the model where useful.

3 Alter one condition. Change a value, representation, requirement or constraint that makes students adapt their reasoning.

4 Predict and test. Ask everyone to explain the expected effect before checking a calculation or completing the altered task.

5 Reset or extend. Use a clean related example for independence, or a further variation for depth; do not force every stage when it adds little.

What this sounds like

“What if this value doubles? Before calculating again, explain which part of your working changes and which part stays the same.”

Classroom example. After a secure worked example, the teacher alters one condition and asks pupils to predict the consequence. Pupils explain which step changes and which relationship stays valid. A clean related task then checks independent setup. The variation deepens the established idea without introducing several new demands at once.

Top tips that make the technique work

Change as little as necessary. A small consequential alteration makes the relationship easier to inspect.

Ask for a prediction first. Students can state what they expect to change and what will remain true.

Use Tick Trick thoughtfully. Compare working with the model to identify the decisions or criteria that earn credit.

Reset when needed. A clean, distinct example tests whether students can start the method independently.

Extend after security. A demanding variation is more useful when the basic method is sufficiently understood.

Do and avoid

Do	Avoid
Preserve the underlying concept.	Changing several unfamiliar demands simultaneously.
Ask for the reason the answer changes.	Using extension while the original decision remains insecure.

Further reading

Practitioner guidance; teacher language above is an illustrative BSCS adaptation.

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